Outbreak Detectives

What is a superbug — and why should you care?

Welcome to our new unit. Before we learn what bacteria are, let's look at what happens when they become impossible to kill. Your job today: be a detective.

PART A The Superbug Numbers

Real data from the World Health Organization and CDC (2021–2023): *• 4.7 million deaths worldwide were linked to antibiotic-resistant bacteria in 2021. • 1 in 6 lab-confirmed bacterial infections worldwide were resistant to antibiotics in 2023. • 9.7 million infections per year in the U.S. are caused by drug-resistant germs, with ~50,800 deaths. • MRSA — a drug-resistant staph bacterium — spreads easily in locker rooms through shared towels, gear, and turf burns.*

1. Look at the number 4.7 million. Put that in perspective: how many times would that fill your school's gym? Make a reasonable estimate and show your math.

2. What does "antibiotic-resistant" mean to you right now, before we've studied it? Write your best guess.

3. MRSA spreads through shared towels and turf burns. As a volleyball player, what are TWO specific moments in a typical practice or game where you could be at risk?

PART B Our Driving Question

Our Unit Question: *Why are bacterial diseases coming back — and how can we stop the next outbreak?*

4. Read the unit question above. Write down THREE questions this makes you want to know the answer to. These will go on our Driving Question Board.

Question 1:

Question 2:

Question 3:

PART C — Unit Map — What's Coming

Look at the unit map on your unit page. For each stop, write one question you're curious about.

Stop	Topic	One thing I'm curious about
1	Meet the Microbes	
2	Plague! How Disease Spreads	
3	Survival of the Fittest	
4	Superbugs	
5	Your Home Team	
6	Team Bacteria	
7	Final Mission	

PART D — Key Words — First Impressions

Before we define these words, write what you THINK each one means. We'll come back to fix these at the end of the unit.

Word	What I think it means NOW	What it actually means (fill in later)
Bacteria		
Antibiotic		
Resistant		
Dormant		
Microbiome		
Natural selection		

■ **Exit Ticket — Driving Question Board** Write your BEST question about bacteria or disease in one sentence. Make it a real question you actually want to know the answer to.

Adapted for 7th grade from New Visions for Public Schools, Humans vs. Bacteria (Biology Unit 2), CC BY-NC-SA 4.0. Data: WHO 2023 · CDC 2023 · Unit page: yscabarney.github.io/MS_SCI_ADV_26/humans-vs-bacteria.html